LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	40
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Explanation of various types of actuators series and parallel and combine.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Explanation of various types of actuators series and parallel and combine.
5. Specific Objectives	14.2.1.1 Trainee will be able to explain the functions of AFCS.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of AFCS actuator schematics; Instructor supervision required for safety regarding hydraulic/electrical systems.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an actuator in an AFCS?
(b) Define series and parallel configurations in hydraulic circuits.
(c) How does actuator failure affect flight control surfaces?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Series actuator configuration analysis • Parallel actuator configuration analysis • Combined series-parallel actuator systems • Failure modes in series arrangements • Failure modes in parallel arrangements	L & Ds	PPT, IB	15 min
(b)	• Hydraulic pressure distribution in series • Flow rate characteristics in parallel • Redundancy benefits of parallel systems • Load sharing in combined configurations • System response time comparisons	L & Ds	PPT, PDF	20 min
(c)	• Application of series actuators in AFCS • Application of parallel actuators in AFCS • Integration of combined systems in AFCS • Maintenance considerations for configurations • Troubleshooting common actuator faults	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference in failure impact between series and parallel actuator systems.
(b) Describe how combined series-parallel configurations enhance AFCS reliability.
(c) Identify key maintenance checks for parallel actuator redundancy.

12. Home Assignment and Follow Up	Prepare a brief note on how actuator configurations in this lesson relate to the stability and control augmentation systems discussed in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Explanation of stability and control augmentation system, attitude base stabilization system.	Refer to the master syllabus reference list	LP 41

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min